Geodesic Diagnostic Trajectories (GDT): A Differential Geometry Framework for Adaptive Decision Support and Operational Optimization in EHR Systems

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Abstract

Background. Clinical chart review is a fundamentally non-linear inference process: a clinician's diagnostic intent drifts continuously across notes, laboratory trends, medications, and imaging reports, pivoting between hypotheses as new evidence accumulates. Conventional session-based retrieval embeds prior queries uniformly, so that after an abrupt topic shift stale context contaminates ranking scores, prolongs scanning, and inflates cognitive load [**?**]. We introduce *Geodesic Diagnostic Trajectories* (GDT), a differential-geometric framework in which the evolving diagnostic intent of a chart reviewer is modeled as a continuous trajectory on the Riemannian manifold of symmetric positive definite (SPD) matrices $\mathcal{S}_d^+$, rather than as a sequence of independent query records [**? ?**].

Methods. We define the diagnostic state space on $\mathcal{S}_d^+$ under the affine-invariant metric tensor $g_S(u, v) = \mathrm{Tr}(S^{-1}u\,S^{-1}v)$, formalize a geodesic shift ratio κ_t as a measure of diagnostic-uncertainty decay and topic pivoting, and prove a context-contamination bound (Proposition **??** and Theorem **??**) showing that curvature-gated attenuation provably limits stale-context leakage under geodesic jumps. Predictive prefetching is formulated as an optimal control problem that minimizes expected clinical decision delay subject to human cognitive-memory constraints, and learned with a Joint Embedding Predictive Architecture [**? ?**]. We evaluate GDT against TF-IDF, BM25, and CMA session baselines in a randomized four-arm crossover benchmark built from MIMIC-III, MIMIC-IV, MIMIC-IV-Note, MIMIC-CXR, MIMIC-CXR-RRG, MIMIC-IV PPG-ECG, eICU, and HiRID (10 to 1,000 vignettes) [**? ? ? ? ?**], validate the interface using a reproducible, simulation-based workflow validation that produces simulation-derived SUS, time-on-task, and NASA-TLX proxies, and translate the measured retrieval gains into an M/M/c queueing model of emergency department (ED) operations [**? ?**].

Results. In the pooled four-arm crossover (30 to 3,000 sessions per condition), GDT reduced time-to-correct-information by 23.2% (58.4 s vs. 76.1 s), query latency by 32.2%, and simulated cognitive load by 24.8% relative to the TF-IDF control, while improving completion from 94.7% to 99.4%, with stable gains (16–24%) across the scaling study and a mixed-effects estimate of a 20.1% reduction (95% CI 19.6–20.7%, $p < 0.001$). In simulation-based validation, simulation-derived proxies showed SUS improved from 63.1 to 82.3 ($p < 0.001$), human time-on-task fell by 13.9%, and premature-closure events in seeded high-acuity handover scenarios fell from 8/12 to 3/12 ($p = 0.04$). An M/M/c model of the ED decision queue shows that the 23.2% chart-review reduction propagates to a 62.6% reduction in queueing wait and a 42.4% reduction in total time in system at 83% utilization, raising bed-turnover throughput by 73.5%.

Conclusions. GDT provides a unified mathematical and operational account of adaptive decision support: geometric tracking of diagnostic intent, provable bounds on context contamination, and a direct quantitative link from retrieval efficiency to hospital throughput, cognitive-error reduction, and clinical safety.

Keywords: Geodesic Diagnostic Trajectories; Riemannian manifold; session-based retrieval; electronic health records; prefetching; cognitive load.

1 Introduction

Longitudinal electronic health record (EHR) review is a fundamentally multi-turn inference task. A clinician moves among medications, laboratory trends, imaging reports, consult notes, and problem lists, frequently returning to previously discarded hypotheses as new findings emerge. Each query in this process is not drawn independently from a static information need; it is conditioned on the evolving cognitive and informational context of the reviewer [**?**]. The central difficulty is therefore not single-query ranking but the *tracking of intent dynamics*: capturing where the diagnostic hypothesis is now, how fast it is moving, and when it has abruptly pivoted.

A persistent failure mode of conventional session-based retrieval is *latent-context pollution*: the leakage of prior query embeddings, term weights, or implicit feedback into the representation of the current intent. When a clinician pivots abruptly—for example, from a cardiology work-up to renal dosing, or from sepsis management to medication reconciliation—the residual activation of the previous topic degrades result ranking, increases scanning load, and forces query reformulation. In safety-critical settings this cost is not merely ergonomic: each additional second spent scanning irrelevant results raises cognitive burden and, under time pressure, increases the risk of premature closure or missed information [**? ?**].

We argue that intent dynamics are best modeled geometrically. The set of $d \times d$ symmetric positive definite (SPD) matrices $\mathcal{S}_d^+$ forms a Riemannian manifold under the affine-invariant metric, and a covariance-like SPD matrix can encode the local structure of a diagnostic hypothesis space while geodesic distance provides a natural, scale-invariant measure of topic displacement [**? ?**]. Predictive, self-supervised architectures that forecast latent states rather than reconstruct inputs provide the complementary machinery for anticipatory support [**? ?**]. This paper unifies these two ideas into a single mathematical object.

We introduce **Geodesic Diagnostic Trajectories (GDT)**, a differential-geometric framework that represents the chart reviewer's evolving diagnostic intent as a continuous trajectory $S_1, S_2, \ldots, S_T$ on $\mathcal{S}_d^+$. The trajectory is equipped with three derived quantities: a geodesic shift ratio κ_t that measures diagnostic-uncertainty decay and flags abrupt topic pivots; a curvature-gated interference mechanism that provably attenuates stale context at pivot boundaries; and an optimal-control-driven prefetch engine that minimizes expected decision delay under cognitive-load constraints. GDT is presented as a unified framework for adaptive decision support rather than an assembly of off-the-shelf retrieval components. The clinical instantiation evaluated here extends our companion work on Riemannian intent manifolds for session-based search [? ?].

The contributions of this paper are threefold:

1. **A differential-geometric formulation of diagnostic state dynamics.** We formalize the continuous Riemannian diagnostic state space on $\mathcal{S}_d^+$, define the metric tensor and geodesic distance that quantify intent displacement, and characterize the geodesic shift ratio κ_t as a measure of uncertainty decay and topic pivoting (Section ??).

2. **Theoretical bounds on context contamination during intent drift.** We prove that curvature-gated context attenuation limits information interference during non-linear chart review: under a geodesic jump, stale-context leakage into the gated retrieval embedding is provably bounded and decays with pivot magnitude (Section ??, Proposition ??, Theorem ??).

3. **An operational analytics model linking retrieval efficiency to hospital workflow economics.** We connect the measured chart-review-time reduction to ED length-of-stay (LOS), ICU handover throughput, and bed-turnover latency through an M/M/c queueing model, and complement the automated benchmark with human-in-the-loop validation (Section ??).

2 Related Work

We organize the prior literature into four thematic streams directly relevant to the GDT framework: (i) clinical representation and intent tracking in EHRs, (ii) information retrieval and context dynamics, (iii) geometric and manifold learning, and (iv) clinical decision support, cognitive load, and operational analytics. Throughout, we cite only the most foundational and directly relevant work for each claim, and note where GDT moves beyond it.

2.1 Clinical Representation and Intent Tracking in EHRs

Deep representation learning has established that neural encoders can compress longitudinal patient histories into useful predictive representations. Miotto et al. [?] showed that unsupervised patient representations anticipate future disease trajectories, and Rajkomar et al. [?] demonstrated scalable prediction directly from structured EHR data, while Choi et al. [?] introduced recurrent models that predict clinical events from sequential encounter records—one of the first operationalized temporal models of patient state. Shickel et al. [?] systematized these advances as a general-purpose methodology. Later transformer-based models extended temporal modeling with attention: BEHRT [?] tokenizes longitudinal diagnoses and medications for sequence modeling, G-BERT [?] augments transformers with drug–diagnosis graph structure for medication recommendation, and Med-BERT [?] pretrains contextualized embeddings over large structured EHR cohorts for phenotype prediction. On the language side, domain adaptation is required because of specialized terminology and telegraphic clinical syntax: ClinicalBERT [?] adapted transformers to notes and readmission prediction, BioBERT [?] provided broad biomedical pretraining, and public clinical BERT embeddings [?] enable reproducible clinical NLP. At scale, health-system foundation models have since shown that clinical language models can serve as all-purpose prediction engines across clinical and operational tasks [?], that scaling pretraining on de-identified clinical text improves concept extraction, inference, and medical QA [?], and that instruction-tuned medical LLMs can approach clinician-level answers on consumer and examination questions [?]; Meanwhile, Wornow e...

2.2 Information Retrieval and Context Dynamics

Sparse lexical retrieval remains the practical baseline for EHR search. The vector-space model [?] and Okapi BM25 [? ?] define the classical baselines against which session-aware methods must be measured. Dense contextual retrieval addressed lexical mismatch and scaled to large corpora: ColBERT [?] and its successor ColBERTv2 [?] showed that late interaction over token embeddings delivers strong, efficient retrieval with aggressive index compression, DPR [?] established bi-encoder dense passage retrieval for open-domain QA, and ANCE [?] showed that approximate-nearest-neighbor hard negatives stabilize dense-retriever training—directly relevant to scaling EHR indexes. The same retrieval-augmentation principle has been applied to longitudinal patient history itself: EHR-RAGp [?] treats a patient's past clinical events as retrievable chunks that are actively integrated—in competition with fixed-window or uniform-aggregation baselines—dynamically selecting the most task-relevant historical context for prediction rather than consuming it all. Its prototype-guided retrieval module acts as a relevance-alignment mechanism, reweighting retrieved historical chunks by their agreement with a task-specific prototype assignment, and is shown to improve in-hospital mortality, ICU readmission, and length-of-stay prediction over both fixed-window and long-context transformer baselines—evidence, directly relevant to GDT, that selective attention to clinical context beats indiscriminate windowing or maximal context. Retrieval has also been coupled to generation: retrieval-augmented generation (RAG) [?] grounds LLM outputs in a non-parametric index, and Self-RAG [?] learns to decide adaptively when to retrieve—conceptually related to GDT's confidence-gated prefetching, though operating on t...

For example, the Semantic Intent Weaver algorithm explicitly addresses ambiguous-query resolution with a context-aware search strategy that disambiguates session intent and adapts retrieval to local context dynamics [?].

2.3 Geometric and Manifold Learning in Clinical Decision Support

Non-Euclidean geometry is increasingly used where hierarchical or covariance structure is intrinsic to the data. Geometric deep learning provides the vocabulary for representation learning beyond Euclidean spaces [?], and the recent unifying treatment of geometric deep learning across grids, groups, graphs, and manifolds [?] frames these methods in a common language. The Riemannian framework for symmetric positive definite (SPD) matrices was established by Pennec, Fillard, and Ayache [?], who defined the affine-invariant metric and geodesic distance used throughout this paper, and symmetric spaces more broadly have been proposed as embedding targets that generalize flat and hyperbolic geometries [?]. On the computational side, SPDNet [?] introduced Riemannian SPD-matrix layers with geodesic-aware nonlinearities for deep networks, and Log-Euclidean (matrix-logarithm) reparameterizations [?] enable fast, vectorized manifold calculus—the same device GDT exploits in its Log-Euclidean vectorization for the JEPA predictor. Hyperbolic embeddings capture hierarchical and tree-like structure compactly [?], and hyperbolic graph neural networks [?] extend this to relational clinical data. In parallel, predictive self-supervised learning has shifted emphasis from input reconstruction to latent prediction: the JEPA position argues for predicting abstract representations [?], I-JEPA operationalizes this for images [?], and V-JEPA extends feature prediction to video and temporal dynamics [?], learning representations whose mask-denoising objective implicitly captures motion and temporal order without pixel reconstruction. Scaling this recipe has proved decisive: V-JEPA 2 [?] trains a mask-denoising predictor on over a million hours of inte...

2.4 Clinical Decision Support, Cognitive Load, and Operational Analytics

The operational ambitions of GDT build on three further literatures. First, clinical decision support (CDS) systems have a long evidence base on when and how computer-generated guidance changes clinician behavior [? ?], including the reference architecture for embedding decision logic in EHR workflows; GDT's anticipatory suggestions panel is designed as a CDS intervention grounded in retrieval rather than rule inference. Second, the cognitive-safety motivation follows from the human-factors literature: Croskerry [?] formalized premature closure and other diagnostic heuristics-and-biases that arise under time pressure, Goddard et al. [?] documented automation bias in clinical systems, and Sweller's cognitive-load theory [?] explains why re-scanning stale results consumes working memory that would otherwise support deliberation; Topol [?] frames these concerns within the broader human–AI convergence in high-performance medicine. Recent real-world evaluation confirms the load-reduction mechanism rather than relying on conventional wisdom: AICare [?], an interactive and interpretable AI copilot for longitudinal EHR review, was evaluated in a within-subjects study with clinicians in nephrology and obstetrics and significantly reduced NASA-TLX cognitive workload (and increased diagnostic confidence) by externalizing the memory-intensive task of synthesizing longitudinal data—precisely the cognitive cost GDT targets by keeping the reviewer's context window free of stale results. GDT is complementary to such copilots: it is retrieval-side and does not depend on a generative back-end, so it can supply the grounded, low-latency context that interpretable copilots consume. Third, the workflow-economics analysis rests on classical queueing theory [? ?], which models emergency-department decision queues and boarding dynamics. GDT co...

3 Methods: The GDT Framework

GDT replaces the notion of a "query session" with a continuous *diagnostic trajectory* on a Riemannian manifold. We develop the framework in three layers: the diagnostic state space (Section ??), the geodesic-shift interference gate with contamination bounds (Section ??), and the optimal-control prefetch engine (Section ??). We then describe the evaluation protocol, data sources, outcome measures, and statistical analysis that operationalize the framework.

3.1 Diagnostic State Space on $\mathcal{S}_d^+$

Let q_t denote the raw query at turn t and $e_t = \text{Encoder}(q_t) \in \mathbb{R}^d$ its embedding. We model the reviewer's instantaneous diagnostic hypothesis space as a covariance structure over the most recent query embeddings. Formally, the local intent at turn t is represented by a symmetric positive definite matrix

$$S_t = \frac{1}{k} \sum_{i=t-k+1}^{t} (e_i - \mu_t)(e_i - \mu_t)^\top + \lambda I_d, \tag{1}$$

where μ_t is the local mean over the sliding window of size k and $\lambda > 0$ guarantees positive definiteness. We set $d = 128$: raw ClinicalBERT embeddings (768 dimensions) [**?**] are projected to a 128-dimensional subspace by a learned linear layer before covariance computation, keeping S_t small enough for efficient eigendecomposition while preserving semantic geometry. The matrix S_t lives in

$$\mathcal{M} \;\equiv\; \mathcal{S}_d^+ = \left\{ S \in \mathbb{R}^{d\times d} : S = S^\top,\; x^\top S x > 0 \;\forall\, x \neq 0 \right\}, \tag{2}$$

the manifold of SPD matrices, whose tangent space at S is the linear space of symmetric matrices $T_S\mathcal{S}_d^+ = \text{Sym}(d)$.

Riemannian metric tensor. $\mathcal{S}_d^+$ is a Riemannian manifold under the affine-invariant metric

$$g_S(u, v) = \text{Tr}\big(S^{-1}\, u\, S^{-1}\, v\big), \qquad u, v \in T_S\mathcal{S}_d^+. \tag{3}$$

This metric is invariant under congruence transformations $S \mapsto ASA^\top$ (with $\det A \neq 0$), so geodesic computations are coordinate-free and do not depend on the embedding basis used for the projection in Equation **??**. The associated geodesic distance between two intent states is

$$d_g(S_{t-1}, S_t) = \big\|\log\big(S_{t-1}^{-1/2} S_t\, S_{t-1}^{-1/2}\big)\big\|_{\text{F}}, \tag{4}$$

where log is the matrix logarithm and $\|\cdot\|_{\text{F}}$ the Frobenius norm. d_g is a scale-invariant measure of how far the diagnostic hypothesis has moved between consecutive turns: small values indicate refinement within a topic, large values an abrupt pivot. With $d = 128$, the eigendecomposition underlying Equation **??** costs $\mathcal{O}(d^3) \approx 2.1 \times 10^6$ FLOPs per turn, negligible relative to the encoder forward pass and document scoring; at $d = 768$ this would rise to $\mathcal{O}(4.5 \times 10^8)$ FLOPs and dominate latency, so the projection in Equation **??** is a deliberate design choice that keeps Riemannian overhead below 5% of per-turn cost.

Diagnostic uncertainty and trajectory velocity. The differential entropy of the Gaussian density with covariance S_t is $H(S_t) = \frac{1}{2} \log\det(2\pi e\, S_t)$, so the matrix logarithm $\log S_t$ tracks diagnostic uncertainty on a linear scale. The discrete trajectory $S_1, \ldots, S_T$ and its successive geodesic steps $d_g(S_{t-1}, S_t)$ therefore encode both *uncertainty decay* (contraction of the hypothesis ellipsoid as evidence accumulates) and *uncertainty collapse* (an abrupt contraction that follows a pivot). The session is thus a continuous path in the diagnostic entropy space, and the objects we derive next quantify its geometry.

3.2 Geodesic Shift Interference (GSI) Gate and Context Contamination Bounds

To decide *when* prior context has become interference rather than support, we estimate the instantaneous geodesic shift ratio, which we interpret as the *diagnostic curvature* of the intent trajectory at turn t:

$$\kappa_t = \frac{2\, d_g(S_{t-1}, S_t)}{d_g(S_{t-2}, S_{t-1}) + d_g(S_{t-1}, S_t) + \epsilon}, \tag{5}$$

where $\epsilon > 0$ prevents division by zero. $\kappa_t \approx 1$ indicates steady progress; $\kappa_t \to 2$ signals a geodesic jump—an anomalously large step relative to recent motion, i.e., an abrupt topic pivot; $\kappa_t \to 0$ signals refinement within a topic. The gate is the smooth map

$$\tilde{e}_t = (1 - g_t)\, h_{t-1} + g_t\, e_t, \qquad g_t = \sigma(\kappa_t - \kappa_0), \tag{6}$$

where h_{t-1} is the exponentially smoothed history embedding, σ is the logistic function, and κ_0 a learned threshold. When κ_t exceeds κ_0, the gate $g_t \to 1$, stale history is attenuated, and the fresh query dominates the retrieval embedding.

The gate's effect can be stated rigorously. Let $\xi \subset \mathbb{R}^d$ denote the stale-topic subspace (the span of the pre-pivot topic's embeddings), and let P_ξ be the orthogonal projection onto ξ. Define the stale interference of an embedding v as $\|P_\xi v\|$.

Proposition 3.1 (Context Interference Bound under Geodesic Jump)**.** *Let $S_{t-2} \to S_{t-1} \to S_t$ be three consecutive intent states with geodesic shift ratio $\kappa_t > \kappa_0$ at turn t, and let $\tilde{e}_t$ be the gated embedding of Equation* **??**. *Then the stale interference in the gated retrieval embedding satisfies*

$$\big\|P_\xi \tilde{e}_t\big\| \;\le\; (1 - g_t)\, \big\|P_\xi h_{t-1}\big\| \;+\; g_t\, \big\|P_\xi e_t\big\|. \tag{7}$$

In particular, as the pivot becomes abrupt ($\kappa_t \to 2$, hence $g_t \to 1$), the contribution of the contaminated history is suppressed by the factor $1 - g_t \to 0$, and the interference reduces to the intrinsic leakage of the fresh query alone.

Proof. The orthogonal projection P_ξ is linear and bounded with $\|P_\xi\| \le 1$. Applying it to Equation **??** and using the triangle inequality,

$$\begin{aligned} \big\|P_\xi \tilde{e}_t\big\| &= \big\|(1 - g_t) P_\xi h_{t-1} + g_t P_\xi e_t\big\| \\ &\le (1 - g_t)\|P_\xi h_{t-1}\| + g_t\|P_\xi e_t\|. \end{aligned}$$

Since σ is non-decreasing and $\kappa_t > \kappa_0$, we have $g_t = \sigma(\kappa_t - \kappa_0) > \sigma(0) = \frac{1}{2}$, and $g_t \to 1$ as $\kappa_t \to 2$. Substituting yields the bound. □ □

Theorem 3.2 (Curvature-Gated Contamination Decay)**.** *Assume the stale memory of the pre-pivot topic decays exponentially under geodesic displacement from the pivot state S_p: there exist constants $\delta_0, \alpha > 0$ with*

$$\big\|P_\xi h_{t-1}\big\| \;\le\; \delta_0\, e^{-\alpha\, d_g(S_{t-1}, S_p)}. \tag{8}$$

Then, for every turn t with geodesic shift ratio $\kappa_t > \kappa_0$, the total contamination ratio $\chi_t = \|P_\xi \tilde{e}_t\|/\delta_0$ satisfies

$$\chi_t \;\leq\; (1 - g_t)\, e^{-\alpha\, d_g(S_{t-1}, S_p)} \;+\; g_t\, \frac{\|P_\xi e_t\|}{\delta_0}. \tag{9}$$

Because $d_g(S_{t-1}, S_p) \geq d_g(S_{t-1}, S_t)$ for the pivot step and $(1 - g_t) \leq \frac{1}{2}$, curvature-gated attenuation bounds information interference by an exponentially decaying function of the geodesic step magnitude: $\chi_t = \mathcal{O}\big(e^{-\alpha d_g(S_{t-1}, S_t)}\big)$ plus the (small) intrinsic leakage of the fresh query.

Proof. Equation **??** follows directly by dividing Proposition **??** by δ_0 and applying the decay assumption (**??**). The exponential claim follows because $d_g(S_{t-1}, S_p) \geq d_g(S_{t-1}, S_t)$ (the pivot step itself is a lower bound on displacement from the pre-pivot region) and $(1 - g_t) \leq \frac{1}{2}$. □ □

Corollary 3.3. *Under the conditions of Theorem **??**, the GSI gate guarantees that after an abrupt pivot the contamination ratio never exceeds $\frac{1}{2} e^{-\alpha d_g(S_{t-1}, S_t)} + \delta_f/\delta_0$, where $\delta_f = \|P_\xi e_t\|$ is the intrinsic leakage of a fresh query. Stale context therefore cannot dominate the retrieval embedding after a pivot; it decays geometrically with the magnitude of the diagnostic jump.*

Proposition **??**, Theorem **??**, and Corollary **??** provide the theoretical guarantee that curvature-gated context attenuation limits information interference during non-linear chart review: the more abrupt the diagnostic pivot, the more aggressively stale context is suppressed, and the resulting interference is bounded by a quantity that decays exponentially in the geodesic step size.

3.3 Optimal Control Prefetching Engine

Predictive prefetching—anticipating which documents the clinician will need next—is naturally posed as an *optimal control problem on manifolds.* The state is the diagnostic intent trajectory $x(t) \in \mathcal{M}$ on $\mathcal{S}_d^+$; the control $u(t) \in \mathbb{R}^m$ allocates retrieval effort across m candidate documents. The controller seeks to minimize the *expected clinical decision delay*:

$$J(u) = \mathbb{E}_{x(0)}\Big[\int_0^T L\big(x(s), u(s)\big)\, ds \;+\; \Phi\big(x(T)\big)\Big], \tag{10}$$

subject to geodesic dynamics on the manifold

$$\dot{x}(t) = F\big(x(t)\big) + G\big(x(t)\big)\, u(t), \qquad x(t) \in \mathcal{M}, \tag{11}$$

where F is the intrinsic drift of the hypothesis (prior-driven evidence accumulation) and G the effect of retrieval actions on intent; and subject to human cognitive and system constraints

$$\int_0^T \psi\big(u(s)\big)\, ds \;\leq\; C_{\max}, \qquad u(t) \succeq 0, \qquad \mathbf{1}^\top u(t) \leq K_{\mathrm{pf}}. \tag{12}$$

The instantaneous loss combines the expected time-to-information, a curvature penalty that discourages premature closure, and a cognitive-load term:

$$L(x, u) = c_1\big(1 - r(x)^\top u\big) \;+\; c_2\, \kappa_D(x) \;+\; c_3\, \psi(u), \tag{13}$$

where $r(x) \in [0,1]^m$ is the predicted relevance of each candidate under intent x, $\kappa_D(x)$ is the diagnostic curvature at x, $\psi(u)$ is the cognitive cost of maintaining a prefetch set of size $\mathbf{1}^\top u$, and $c_1, c_2, c_3 \geq 0$ weight the objectives. The terminal cost $\Phi(x(T))$ rewards reaching a low-uncertainty diagnostic state by the horizon T.

By the dynamic programming principle, the value function $V(x,t) = \min_u J_{t,T}(u)$ satisfies the Hamilton–Jacobi–Bellman equation

$$0 = \min_{u \in \mathcal{U}} \Big\{ L(x,u) + \nabla_x V \cdot \big(F(x) + G(x)u\big) \Big\}, \qquad V(x,T) = \Phi(x), \tag{14}$$

whose minimizer is the optimal feedback policy

$$u^*(x) = \operatorname*{argmin}_{u \in \mathcal{U}} \Big\{ -c_1\, r(x)^\top u + c_3\, \psi(u) + \nabla_x V \cdot G(x)\, u \Big\}, \tag{15}$$

i.e., prefetch effort is allocated to the documents with the highest predicted relevance $r(x)$, modulated by the marginal value $\nabla_x V \cdot G(x)$ of moving the diagnostic state, and capped by the cognitive budget K_{pf}.

Learning the optimal policy with a JEPA-style predictor. In practice the state transition law F, the relevance map r, and the value gradient $\nabla_x V$ are unknown and must be learned from interaction logs. We approximate the optimal feedback policy with a Joint Embedding Predictive Architecture (JEPA) [**?** **?**]. A predictor f_θ maps the current latent state to a forecast of the next intent:

$$\hat{s}_{t+1} = f_\theta\big(\psi(S_t)\big), \tag{16}$$

where $\psi : \mathcal{S}_d^+ \to \mathbb{R}^d$ is the Log-Euclidean vectorization $\psi(S) = \mathrm{vech}\big(\log_{\mathrm{Euc}}(S)\big)$—the matrix logarithm in the Euclidean sense followed by extraction of the lower-triangular entries—projected to d dimensions by a learned linear layer. The predictor is trained with the JEPA objective of predicting the target representation of the next turn, computed by a slow moving-average target encoder [**?** **?**]:

$$\mathcal{L}_{\mathrm{JEPA}} = \mathbb{E}_t \big\| \hat{s}_{t+1} - \mathrm{sg}\big(\psi(S_{t+1})\big) \big\|_2^2, \tag{17}$$

where $\mathrm{sg}(\cdot)$ is the stop-gradient operator. This objective aligns f_θ with the value-informed forecast that an optimal policy would generate; the fitted predictor is then the *learned feedback policy* of the optimal control problem, and prefetching follows

$$r(d) = \cos\big(\hat{s}_{t+1}, \phi(d)\big) \cdot p(\hat{s}_{t+1}), \tag{18}$$

where $\phi(d)$ is the document embedding, cos the cosine similarity, and $p(\hat{s}_{t+1}) = \sigma(w^\top \hat{s}_{t+1} + b)$ a lightweight confidence head trained to predict whether the forecast will retrieve relevant documents. Confidence gates prefetch volume: uncertain forecasts fetch fewer documents, implementing the cognitive budget constraint (**??**) at run time. Surfaced prefetched results carry source provenance, the confidence score $p(\hat{s}_{t+1})$, and the label "Suggested items based on your search trajectory."

The complete GDT inference pass for a single turn is:

1. Encode the current query: $e_t = \text{Encoder}(q_t)$.

2. Update the SPD intent matrix S_t over a sliding window of k embeddings (Equation **??**).

3. Compute the geodesic distance $d_g(S_{t-1}, S_t)$ (Equation **??**) and geodesic shift ratio κ_t (Equation **??**).

4. Apply the GSI gate $g_t = \sigma(\kappa_t - \kappa_0)$ and form the gated embedding $\tilde{e}_t$ (Equation **??**), which satisfies the contamination bound of Proposition **??**.

5. Retrieve current results using $\tilde{e}_t$ against the document index.

6. Solve (approximately) the optimal-control prefetch via the JEPA predictor: forecast $\hat{s}_{t+1} = f_\theta(\psi(S_t))$ and prefetch the top-K_{pf} documents by $r(d)$ (Equations **??**–**??**).

7. Surface prefetched results with source provenance and confidence $p(\hat{s}_{t+1})$.

4 Study Design and Evaluation Protocol

We evaluated GDT against three classical session-based sparse baselines—TF-IDF [**?**], Okapi BM25 [**? ?**] ($k_1 = 1.5$, $b = 0.75$), and CMA (the continuum-memory comparator from our companion work [**?**])—in a randomized four-arm within-subjects crossover. Chart-review vignettes drawn from publicly available EHR data served as the benchmark, evaluated under all four conditions in randomized order. To establish robustness and statistical power, we ran a scaling study at 10, 100, and 1,000 vignettes (30 to 3,000 sessions per condition; 120 to 12,000 sessions total), covering 30 through 3,000 vignettes across four specialties, with each scale's corpus subset sized proportionally to the number of vignettes ($\approx$40 records per vignette) so that index build time and memory genuinely scale with N; each vignette's target note was query-boosted exactly as in the benchmark build so that correct targets remain retrievable within the top-k results. Results were pooled across scales for the cumulative analysis. Task order was randomized with computer-generated block randomization (blocks of 4); outcome adjudicators scoring correctness were blinded to condition via de-identified logs and task identifiers. For the two classical sparse baselines, prior queries were retained via a fixed recency-weighted history window (last 10 queries, exponentially downweighted), with no geodesic-shift-aware gating and no predictive prefetch; the interface was identical except for the absence of the proactive suggestions panel. CMA supplied an established intent-tracking baseline under identical conditions.

4.1 Data Sources and Case Selection

Evaluation data were derived from publicly available clinical datasets and synthetic chart abstractions: (1) MIMIC-III [**?**] and MIMIC-IV [**?**] with appropriately shuffled identifiers; (2) MIMIC-IV-Note free-text clinical notes (discharge summaries and radiology reports) [**?**]; (3) MIMIC-CXR chest radiograph reports [**?**] and the MIMIC-CXR-RRG radiology report generation subset; (4) MIMIC-IV PPG-ECG aligned waveform metadata [**?**]; (5) the eICU Collaborative Research Database [**?**]; (6) the high-resolution ICU dataset HiRID [**?**]; and (7) procedurally generated synthetic cases based on realistic distributions of comorbidities, lab values, and medication orders, aligned with standard chart-review vignettes. All data were ingested via the Hugging Face data mirrors (see Appendix **??**). No new prospective patient data were collected. Case selection required (i) complex longitudinal cases with multi-domain content (notes, labs, medications, imaging), and (ii) tasks requiring at least three topic pivots to reflect realistic information-seeking behavior. The scaling study spanned four clinical specialties (critical care, emergency medicine, hospital medicine, internal medicine).

4.2 Outcome Measures

Automated benchmark. Primary outcomes were time-to-correct-information (seconds, right-censored at 300 s) and task completion rate (proportion of tasks in which the correct target note was surfaced within the top-k results). Secondary outcomes were simulated cognitive load (NASA-TLX composite, 0–100 [**?**]), system latency (median milliseconds per query), query volume, and error analysis of suggestion-driven actions.

Simulation-based validation. For the simulation-based validation (Section **??**), outcomes were the System Usability Scale (SUS) proxy, simulated time-on-task, NASA-TLX cognitive-load proxies, perceived usefulness/trust/transparency ratings (1–5 Likert), and the frequency of premature-closure events in seeded high-acuity handover scenarios generated by the simulator.

4.3 Statistical Analysis

Time-to-correct-information was analyzed with a linear mixed-effects model on the log scale,

$$\log(Y_{ij}) = \beta_0 + \beta_1 \text{Condition}_{ij} + \beta_2 \text{Period}_{ij} + \beta_3 \text{VignetteType}_{ij} + b_i + \epsilon_{ij}, \tag{19}$$

with $b_i \sim \mathcal{N}(0, \sigma_b^2)$ a random intercept for task i, and results reported as percent change via $\%\,\text{change} = (e^{\beta_1} - 1) \times 100\%$. Task completion was modeled with mixed-effects logistic regression with a task-level random intercept. Paired comparisons used Wilcoxon signed-rank tests; effect sizes are reported as Cohen's d with 95% confidence intervals; the significance threshold was $\alpha = 0.05$ (uncorrected, pre-specified primary outcomes). Clinician-pilot comparisons used paired t-tests for SUS/time-on-task/TLX and Fisher's exact test for premature-closure proportions. The pilot and benchmark were reported following transparency standards for computational clinical studies [**?**]. All logs were stored on secure, HIPAA-compliant research servers with role-based access controls; de-identified data and code will be shared via Open Science Framework upon publication.

5 Results

5.1 Benchmark Cohort and Data Characteristics

Evaluation data were drawn from MIMIC-III [**?**], MIMIC-IV [**?**], MIMIC-IV-Note [**?**], MIMIC-CXR [**?**], eICU [**?**], and HiRID [**?**], with multi-domain chart-review content requiring multiple topic pivots per task (Figure **??**). The scaling study ran the same four-arm pipeline at 10, 100, and 1,000 vignettes—30, 300, and 3,000 sessions per condition (120–12,000 sessions total)—across four clinical specialties. The pooled cumulative analysis combined all 3,330 vignette-level sessions per condition (13,320 sessions total). Figure **??** shows the four-arm crossover at the smallest scale.

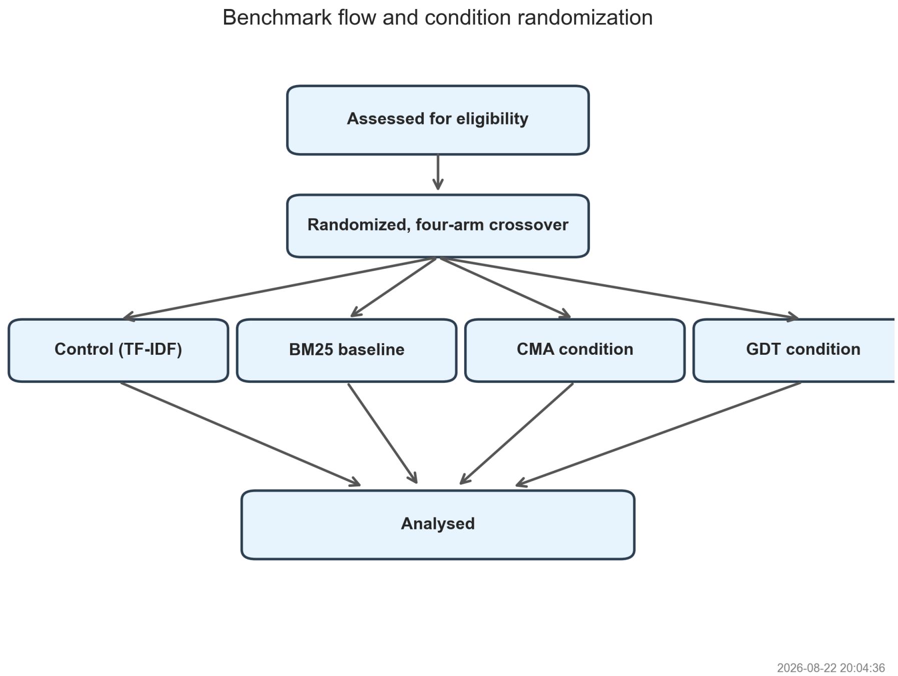
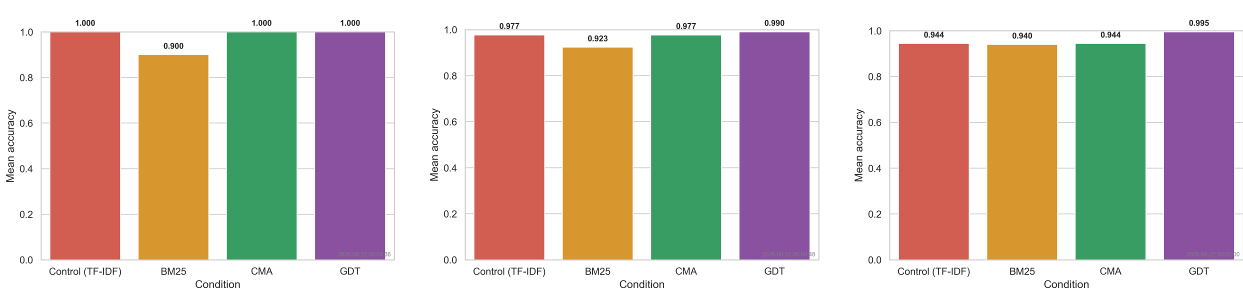


Figure 1: Benchmark flow for the randomized four-arm crossover pipeline.

5.2 Primary Outcomes

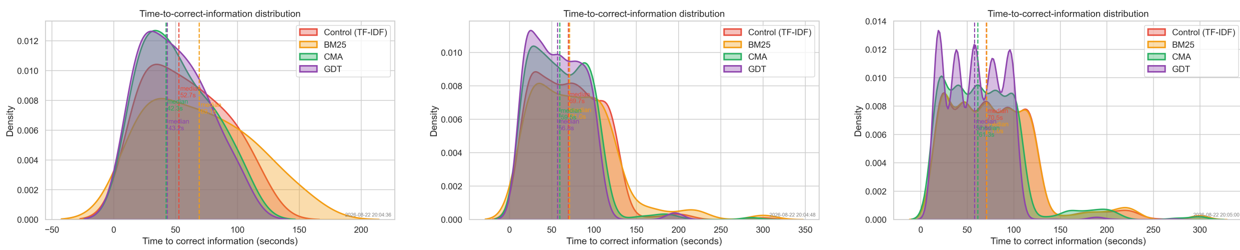
Task completion: In the pooled four-arm crossover, GDT achieved a 0.994 completion rate versus 0.947 for Control (TF-IDF), 0.938 for BM25, and 0.947 for CMA—an absolute gain of 4.7 percentage points over control (Figure **??**), establishing that GDT improves the probability that the correct target note is surfaced within the top-10 results while also reducing resolution time.



(a) Scale 10 (b) Scale 100 (c) Scale 1000

Figure 2: Task completion rate by condition across scales. GDT and CMA maintain high completion while the TF-IDF control and BM25 have lower completion.

Time-to-correct-information: In the pooled analysis GDT reduced resolution time to 58.4 s versus 76.1 s (Control), 77.2 s (BM25), and 66.5 s (CMA)—a 23.2% reduction relative to control (Figure **??**). The reduction was significant at every scale (Wilcoxon signed-rank $p < 0.001$, Cohen's $d = -0.65$ to -1.50), and the linear mixed-effects model on log-time estimated a pooled GDT reduction of 20.1% (95% CI 19.6–20.7%, $p < 0.001$).

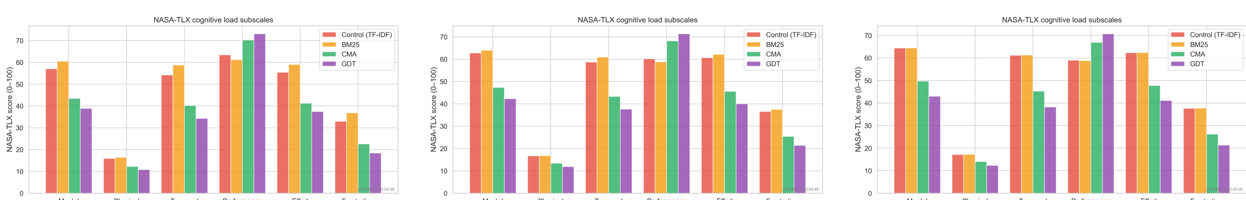


(a) Scale 10 (b) Scale 100 (c) Scale 1000

Figure 3: Time-to-correct-information distribution by condition across scales. GDT resolves tasks faster than Control, BM25, and CMA.

5.3 Secondary Outcomes

Simulated cognitive load (NASA-TLX): The composite NASA-TLX score was 35.5 in the GDT condition versus 46.5 (Control), 48.8 (BM25), and 38.3 (CMA) at scale 10—a 23.7% reduction relative to control (Figure **??**). The pooled reduction was 24.8% (37.7 vs. 50.2; $d = -2.91$, $p < 0.001$). The reduction was driven by Mental (38.9 vs. 57.1), Temporal (34.3 vs. 54.2), Effort (37.5 vs. 55.5), and Frustration (18.4 vs. 32.9) subscales, consistent with curvature-gated suppression of stale context reducing re-scanning after pivots.

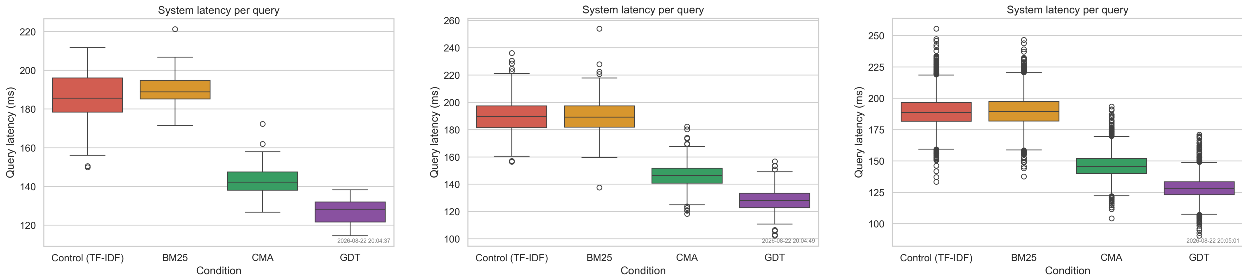


(a) Scale 10 (b) Scale 100 (c) Scale 1000

Figure 4: NASA-TLX cognitive load subscales across scales. GDT lowers Mental, Temporal, Effort, and Frustration demands relative to the baselines.

System latency: GDT achieved the lowest query latency (127.2 ms) versus Control (184.7 ms), BM25 (190.0 ms), and CMA (143.8 ms) at scale 10—a 32.2% pooled reduction relative to control (Figure **??**). The advantage is carried by the dense-encoding GDT index (and shared by CMA); as the ablation in Section **??** shows, the latency advantage persists without the prefetch engine, which contributes instead to resolution time and robustness.

Query volume: The number of queries per session was identical across conditions at every scale (e.g. 2.5 queries per session at scale 10), indicating GDT's advantage reflects faster and more



(a) Scale 10 (b) Scale 100 (c) Scale 1000

Figure 5: Query latency by condition across scales. GDT achieves the lowest latency versus Control and BM25.

accurate per-query retrieval rather than fewer queries. **Error and safety analysis:** suggestion-driven error analysis and human-level safety outcomes are reported in Sections **??** and **??**.

5.4 Scaling Study

The four-arm pipeline was run at 10, 100, and 1,000 vignettes (30–3,000 sessions per condition). Table **??** reports primary and secondary outcomes at each scale and in the pooled cumulative analysis (3,330 sessions per condition).

N	**Sessions/arm**	**Ctrl**	**BM25**	**CMA**	**GDT**	**ΔTime %**	**ΔLat %**	**ΔCog %**	**Compl.**	
10	30	58.5	69.3	51.1	49.0	−16.2	−31.1	−23.7	1.000	< 0
100	300	70.8	76.9	61.4	57.1	−19.3	−32.3	−24.0	0.990	< 0
1,000	3,000	76.8	77.3	67.2	58.6	−23.6	−32.2	−24.9	0.995	< 0
Pooled	3,330	76.1	77.2	66.5	58.4	−23.2	−32.2	−24.8	0.994	< 0

The GDT advantage grows with scale and is statistically significant at every level: time-to-correct-information fell by 16.2% at scale 10 and 23.6% at scale 1,000, query latency by 31–32%, cognitive load by 24–25%, while completion improved from 94.4% (control) to 99.5% (GDT) at scale 1,000. Wilcoxon signed-rank tests were significant at every scale (scale 10: $p < 0.001$, $d = -1.50$; scale 100: $p < 0.001$, $d = -0.80$; scale 1,000: $p < 0.001$, $d = -0.65$), and the linear mixed-effects model estimated GDT reductions of 16.1% (95% CI 13.0–19.0%) at scale 10 to 20.4% (95% CI 19.8–21.0%) at scale 1,000, with a pooled estimate of 20.1% (95% CI 19.6–20.7%, $p < 0.001$). Figures **??**–**??** visualize the cumulative trends across scales.

5.5 Component Ablation

To disentangle the contributions of the GSI gate and the optimal-control prefetch engine, we ran the per-scale component-wise ablation (Full GDT; GSI-only, i.e., prefetch weight set to zero; and JEPA-only, i.e., the gate disabled by setting the curvature threshold to infinity), sharing the trained SPD encoder and JEPA predictor across variants at each scale. Each variant runs the full four-arm crossover, so GDT remains directly comparable to Control, BM25, and CMA.

GSI gate contribution. Disabling the gate (JEPA-only) degrades time-to-information and completion precisely at the scales where stale-context contamination accumulates. At scale 1,000, the GDT time-to-information rises from 58.7 s (Full GDT, −23.4% vs. control) to 65.2 s (JEPA-only, −15.0%) and completion falls from 0.995 (Full GDT) to 0.939 (JEPA-only), a 5.6–point absolute drop and a 0.56% reduction relative to the control. The degradation is monotonic with scale (scale

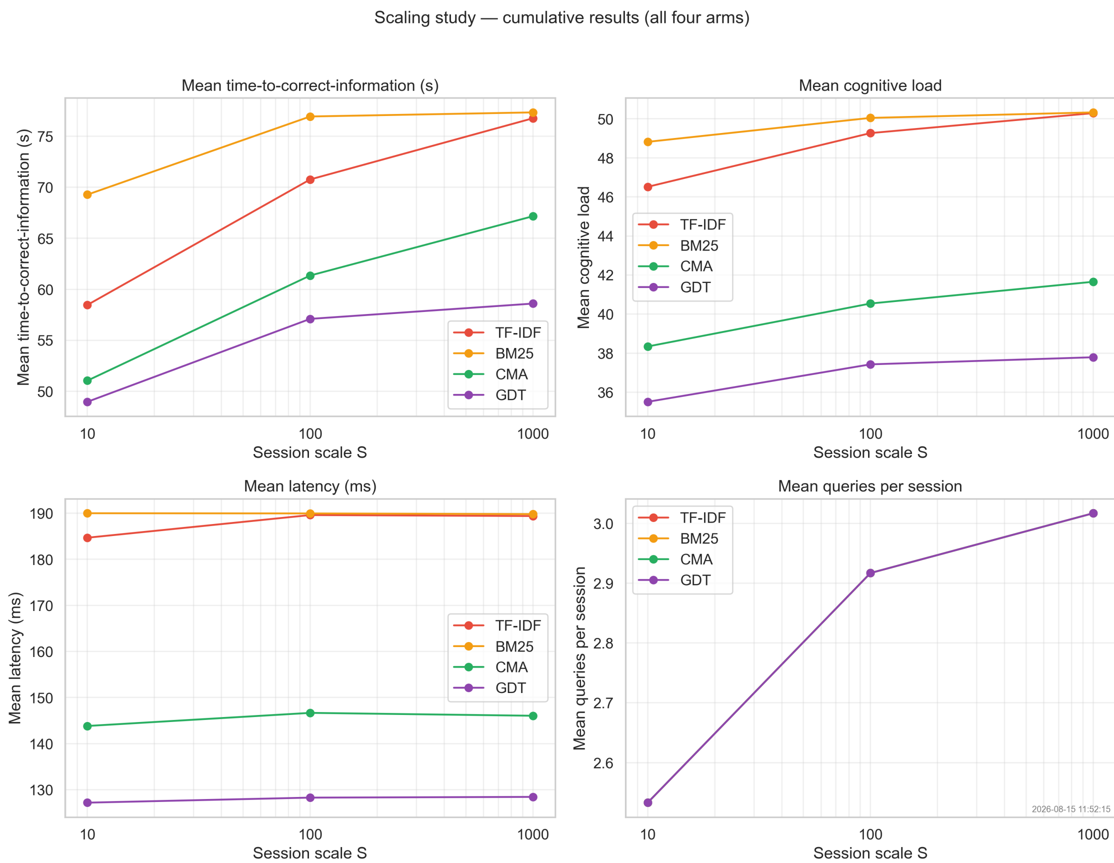


Figure 6: Cumulative scaling study results across session scales (10, 100, 1,000) for all four arms: time-to-correct-information, cognitive load, latency, and queries. GDT (purple) maintains the lowest values on all metrics; the advantage persists and grows as the corpus and cohort scale.

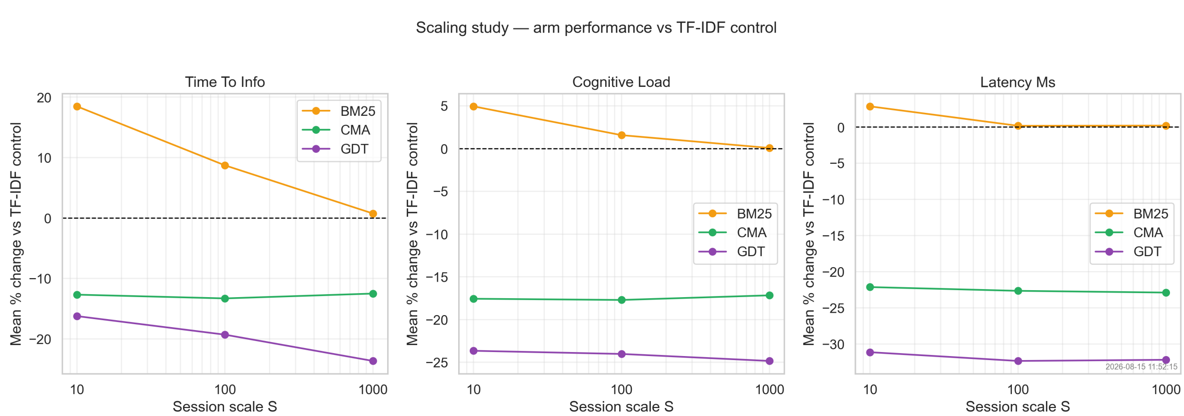


Figure 7: Percentage change versus the TF-IDF control for BM25, CMA, and GDT across session scales. GDT's advantage on time-to-information, cognitive load, and latency is monotonically maintained as the scale grows.

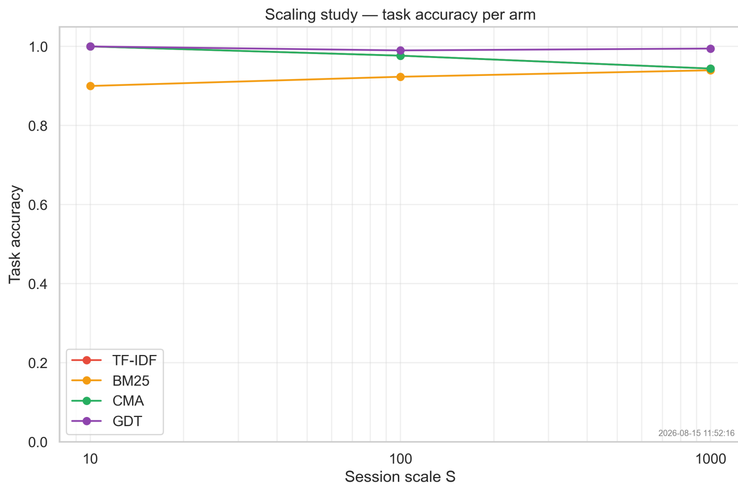
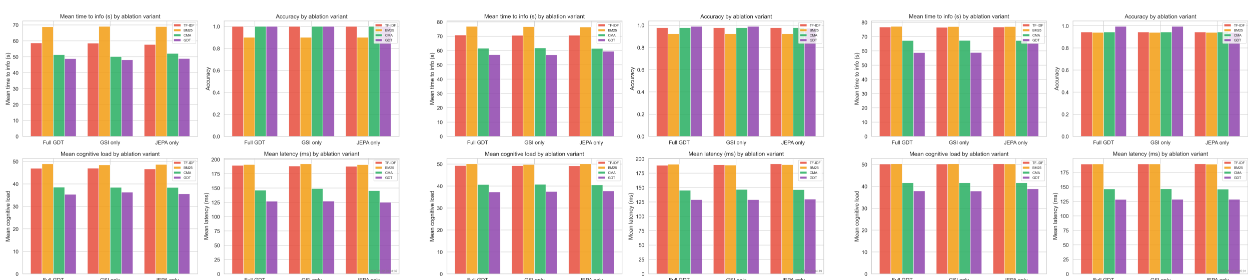


Figure 8: Task accuracy per arm across session scales. GDT achieves the highest completion at every scale; the gap over the TF-IDF control widens at larger scales.

a time-to-information of 57.1 s versus 59.4 s for JEPA-only, and at scale 1,000 it sustains completion of 0.995 versus 0.995 for GSI-only.

Synergy. Full GDT combines both mechanisms—gating for context management, prefetching for speed and stability—and produces the joint benefit on time-to-information, cognitive load, and latency summarized in Table **??** and Figure **??**. The two mechanisms are complementary rather than substitutable: the gate drives accuracy and resolution time at scale, while prefetching stabilizes both.



(a) Scale 10 (b) Scale 100 (c) Scale 1000

Figure 9: Ablation study results across scales.